

Module 1: SQL Fundamentals

The foundation of every SQL query, database operation, and data analytics project starts here.



Introduction

SQL (Structured Query Language) is the standard language used to communicate with relational databases. It allows users to create, store, retrieve, modify, and manage data efficiently.

Whether you want to become a Data Analyst, Business Analyst, Data Engineer, Backend Developer, or Database Administrator, SQL is one of the most important technical skills to learn.



Learning Objectives

After completing this module, you will be able to:

- Understand what databases are
 - Differentiate between DBMS and RDBMS
 - Understand why SQL is used
 - Learn SQL command categories
 - Understand database terminology
 - Write your first SQL query
 - Understand how SQL fits into Data Analytics
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1 What is Data?

Data refers to raw facts, observations, or information.

Examples:

EmployeeID	Name	Salary
101	Kunal	45000
102	Rahul	50000
103	Aman	55000

In this table:

- 101, 102, 103 are data
- Names are data
- Salaries are data

2 What is a Database?

A database is an organized collection of related data stored electronically.

Think of a database as a digital filing cabinet.

Example:

A company stores:

- Employee Information
- Customer Information
- Sales Data
- Product Data

All these records together form a database.

Without Database

Imagine storing employee information in:

- Notebooks
- Excel sheets
- Text files

Problems:

✗ Duplicate records

- ✗ Data inconsistency
 - ✗ Difficult searching
 - ✗ Security issues
 - ✗ Slow updates
-

With Database

Advantages:

- ✓ Fast searching
 - ✓ Easy updates
 - ✓ Security
 - ✓ Backup support
 - ✓ Data consistency
 - ✓ Multi-user access
-

3 What is DBMS?

DBMS stands for Database Management System.

It is software that allows users to create and manage databases.

Responsibilities of DBMS

A DBMS:

- Creates databases
 - Stores data
 - Retrieves data
 - Updates data
 - Deletes data
 - Manages security
 - Handles backups
-

Examples of DBMS

- Microsoft SQL Server
- MySQL
- PostgreSQL
- Oracle Database

- SQLite



What is RDBMS?

RDBMS stands for Relational Database Management System.

An RDBMS stores data in the form of tables.

Example

Employees Table

EmployeeID	Name
101	Kunal
102	Rahul

Departments Table

DepartmentID	DepartmentName
1	HR
2	IT

These tables can be connected using relationships.

Key Features of RDBMS

Tables

Data is stored in rows and columns.

Relationships

Tables can be connected.

Primary Keys

Uniquely identify records.

Foreign Keys

Create relationships between tables.

DBMS vs RDBMS

Feature	DBMS	RDBMS
Data Storage	Files	Tables
Relationships	Limited	Strong
Normalization	Optional	Supported
Security	Basic	Advanced
Examples	File Systems	SQL Server

5 What is SQL?

SQL stands for:

Structured Query Language

It is used to communicate with relational databases.

SQL helps us:

- Create tables
- Insert data
- Retrieve data
- Update data
- Delete data

Example Query

```
SELECT * FROM Employees;
```

Meaning:

- SELECT → Retrieve data
- ◦ → All columns
- FROM → Source table
- Employees → Table name

Output

```
| EmployeeID | Name |  
|-----|-----|  
| 101 | Kunal |
```

| 102 | Rahul |

6 Why Learn SQL?

SQL is one of the most demanded skills in:

- Data Analytics
- Business Intelligence
- Data Engineering
- Software Development
- Data Science

Job Roles Using SQL

Data Analyst

- Extract data
- Analyze trends
- Create reports

Business Analyst

- Generate business insights

Data Engineer

- Build data pipelines

Database Administrator

- Manage databases

Backend Developer

- Connect applications with databases

7 Important Database Terminology

Database

Collection of related data.

Example:

```
```text
CompanyDB
```

---

## Table

Collection of rows and columns.

Example:

```
Employees
```

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## Row (Record)

Single entry in a table.

Example:

EmployeeID	Name
101	Kunal

This is one row.

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## Column (Field)

Specific attribute.

Example:

Name
Kunal
Rahul

Name is a column.

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## Schema

Blueprint of a database.

Defines:

- Tables
- Columns
- Data Types

- Relationships

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## Primary Key

Uniquely identifies a row.

Example:

<b>EmployeeID</b>
101
102

Cannot:

✗ Repeat

✗ Be NULL

---

## Foreign Key

Creates relationship between tables.

Example:

Employee table stores:

DepartmentID

which references

Departments.DepartmentID

---

## SQL Command Categories

SQL commands are divided into 5 categories.

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## DDL (Data Definition Language)

Used to define database structure.

Commands:



```
CREATE
ALTER
DROP
TRUNCATE
```

---

## CREATE

Creates database objects.

```
CREATE TABLE Employees
(
 EmployeeID INT,
 Name VARCHAR(100)
);
```

---

## ALTER

Modifies table structure.

```
ALTER TABLE Employees
ADD Salary INT;
```

---

## DROP

Deletes object permanently.

```
DROP TABLE Employees;
```

---

## TRUNCATE

Deletes all rows.

```
TRUNCATE TABLE Employees;
```

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## DML (Data Manipulation Language)

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Used to manipulate data.

Commands:

```
INSERT
UPDATE
DELETE
```

---

## INSERT

Adds records.

```
INSERT INTO Employees
VALUES
(101, 'Kunal');
```

---

## UPDATE

Modifies data.

```
UPDATE Employees
SET Name='Rahul'
WHERE EmployeeID=101;
```

---

## DELETE

Removes rows.

```
DELETE FROM Employees
WHERE EmployeeID=101;
```

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## DQL (Data Query Language)

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Used for data retrieval.

Command:

```
SELECT
```

Example:

```
SELECT *
FROM Employees;
```

---

## DCL (Data Control Language)

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Controls permissions.

Commands:

```
GRANT
REVOKE
```

---

## TCL (Transaction Control Language)

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Controls transactions.

Commands:

```
COMMIT
ROLLBACK
SAVEPOINT
```



## SQL Query Processing Order

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When SQL executes:

```
SELECT Name
FROM Employees
WHERE Salary > 50000;
```

Database processes:

```
FROM
WHERE
SELECT
```

Actual execution order:

1. FROM
2. WHERE
3. SELECT

Understanding this is important for advanced SQL.

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## 10 First SQL Query

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```
SELECT *
FROM Employees;
```

Breakdown:

Keyword	Meaning
SELECT	Retrieve data
*	All columns
FROM	Source table
Employees	Table name

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## 1 1 SQL in Data Analytics

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SQL is used daily for:

Data Extraction

```
SELECT *
FROM Sales;
```

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Data Cleaning

```
UPDATE Customers
SET City='Delhi'
WHERE City='delhi';
```

---

## Data Aggregation

```
SELECT SUM(Sales)
FROM Orders;
```

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## Reporting

```
SELECT Month,
 SUM(Revenue)
FROM Sales
GROUP BY Month;
```

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# 1 2 Real-World SQL Workflow

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A Data Analyst typically:

### Step 1

Receive business question.

Example:

Which products generated the highest revenue?

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### Step 2

Write SQL query.

```
SELECT ProductName,
 SUM(Revenue)
FROM Sales
GROUP BY ProductName;
```

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### Step 3

Analyze output.

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## Step 4

Build dashboard using:

- Power BI
  - Tableau
  - Excel
- 



## Summary

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In this module, you learned:

- ✓ What data is
  - ✓ What databases are
  - ✓ What DBMS and RDBMS are
  - ✓ What SQL is
  - ✓ Why SQL is important
  - ✓ Database terminology
  - ✓ SQL command categories
  - ✓ Query processing basics
  - ✓ SQL applications in analytics
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## Practice Questions

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### Theory

1. What is a database?
  2. Difference between DBMS and RDBMS?
  3. What is SQL?
  4. What is a Primary Key?
  5. What is a Foreign Key?
  6. Explain DDL commands.
  7. Explain DML commands.
  8. Explain DCL commands.
  9. Explain TCL commands.
  10. Why is SQL important for Data Analysts?
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## Practical

Create a table:

Students

with columns:

StudentID  
StudentName  
Age  
Course

Then:

1. Insert 5 records
2. Display all records
3. Update one record
4. Delete one record



## Next Module

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 Module 2: Database Objects

Topics:

- Database
- Table
- Row
- Column
- Schema
- Keys
- Relationships
- Constraints Introduction